

Kinetic Command: A Real-Time, Gesture-Based System for Gamified Hand Rehabilitation Using MediaPipe and OpenCV

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Abstract—*Kinetic Command* is a real-time, webcam-only rehabilitation system that enhances at-home hand therapy using quantitative motion tracking and gamified feedback. The system addresses three persistent challenges in telerehabilitation: (i) poor at-home adherence due to monotony and discomfort, (ii) limited accessibility of specialized sensing hardware (e.g., instrumented gloves, depth cameras), and (iii) the absence of real-time, objective feedback on movement quality. Built on Google’s MediaPipe Hands for 21-point landmark inference and OpenCV for interactive overlays, the system computes biomechanically meaningful metrics such as joint angles, fingertip distances, and range of motion. It validates repetitions via a deterministic state machine with hold timers and provides multimodal feedback through progress bars, visual cues, and auditory signals. Six representative rehabilitation exercises—hand open/close, wrist flex/extend, finger spread, thumb opposition, wrist deviation, and index tap—are implemented and evaluated on a CPU-only laptop. Results demonstrate near-30.00 FPS responsiveness and over 92% detection accuracy while relying only on a standard webcam. The findings suggest that lightweight computer vision models can deliver engaging, low-cost telerehabilitation experiences with quantitative insights suitable for daily use.

Index Terms—Rehabilitation, Computer Vision, MediaPipe, OpenCV, Pose Estimation, Gamification, Human-Computer Interaction, Telerehabilitation.

I. INTRODUCTION

Hand and wrist rehabilitation is essential following musculoskeletal injuries (e.g., tendon damage, fractures), repetitive strain syndromes (e.g., carpal tunnel), and neurological conditions (e.g., post-stroke hemiparesis). Standard care involves structured, repetitive exercises to rebuild dexterity, range of motion (ROM), strength, and motor control under therapist guidance. However, therapy often extends beyond supervised sessions into at-home practice, where three persistent challenges emerge: (1) *adherence* declines outside the clinic due to monotony, pain, or uncertainty; (2) *accessibility* is constrained by the cost, availability, and setup complexity of specialized sensing hardware; and (3) *feedback* is limited, with patients lacking

objective, real-time indicators of movement quality and progress.

Recent advances in AI-based human pose estimation have reduced the hardware barrier for motion analysis. Lightweight, CPU-efficient frameworks such as MediaPipe Hands enable robust 21-landmark hand tracking from ordinary RGB webcams in real time. When combined with a carefully designed feedback loop—clear instructions, progress visualization, adaptive thresholds, and immediate audio confirmation—these models can transform repetitive drills into measurable, motivating micro-tasks.

Kinetic Command operationalizes this insight. The system estimates per-exercise quantitative metrics (e.g., 3D fingertip-wrist distances, inter-finger spread, and joint angles), applies a deterministic state machine with dwell-time validation to suppress jitter-induced false positives, and provides gamified feedback to sustain engagement. Distinct from binary “pose classification,” our objective is *continuous, interpretable measurement* aligned with clinical goals (ROM, posture, smoothness), delivered via accessible, non-intrusive technology.

A. Contributions

This project makes the following contributions:

- **Webcam-only, quantitative telerehabilitation:** A fully software-based pipeline that tracks six hand exercises using MediaPipe Hands and renders interactive feedback in OpenCV, with no proprietary sensors.
- **Biomechanics-aligned metrics:** Continuous, interpretable features (joint angles, Euclidean distances, ROM) rather than categorical gestures, aligning more closely with therapeutic assessment.
- **Robust repetition validation:** A state machine with hold timers and threshold hysteresis to mitigate landmark jitter and transient occlusion, improving repetition detection fidelity.
- **Gamified feedback loop:** Progress bars, text guidance, and audio cues that increase clarity and motivation in otherwise monotonous routines.
- **Empirical evaluation:** Throughput, latency, and accuracy characterization on CPU-only laptops, demonstrating near-30.00 FPS responsiveness and high recognition correctness across exercises.

B. Organization

Section II surveys related work across sensorized gloves, depth cameras, and modern vision-based approaches. Section III details methodology, including the system pipeline, the mathematical formulation of metrics, and the repetition state machine. Sections IV and V present the experimental setup and results. Section VII provides a multidimensional discussion (statistical interpretation, clinical relevance, gamification theory) and Section VIII outlines limitations and future work before concluding in Section IX.

II. LITERATURE REVIEW

Research on digital rehabilitation spans multiple sensing paradigms—from manual goniometry and wearable instrumentation to depth cameras and AI-based markerless pose estimation. Here we review representative approaches, focusing on technical trade-offs, clinical alignment, and suitability for home deployment.

A. Manual and Wearable Measurement

The clinical gold standard for range-of-motion assessment is manual goniometry conducted by trained therapists. While accurate in session, it is impractical for unsupervised daily monitoring and does not scale to continuous feedback outside the clinic. To bridge this gap, early digital systems employed *wearable* sensors. Instrumented gloves (e.g., bend or fiber-optic goniometers) and IMUs (accelerometers and gyroscopes) can capture finger flexion, wrist posture, and movement dynamics with high fidelity. Studies report precise kinematics and rich signals for modeling, enabling detailed analysis of motion quality and tremor, as well as closed-loop training aids.

However, the benefits of wearables come with costs: device expense, fit/hygiene concerns (especially when shared), donning difficulty (notably for impaired hands), calibration drift, and maintenance overhead. In addition, wired or battery-powered designs may constrain natural movement and reduce willingness to use the devices at home over extended periods. For population-scale telerehabilitation, these factors limit adoption and long-term adherence despite strong measurement quality.

B. Depth Cameras and Proprietary Sensors

Markerless depth-sensing devices (e.g., Microsoft Kinect, Leap Motion) popularized affordable 3D tracking in the 2010s. Kinect’s skeleton tracking and Leap Motion’s dedicated hand tracking enabled body and hand pose reconstruction with good accuracy under controlled conditions. They offered a better user experience than wearables—no contact, faster setup, and relatively robust 3D geometry.

Limitations nonetheless remain. Proprietary hardware introduces dependencies on supply chains, drivers, and platform support; with discontinued products, legacy projects become difficult to sustain. Interaction volumes are constrained (e.g., Leap’s optimal field of view and range), and some sensors require specific mounting or lighting

conditions. For long-term home use, device availability and OS compatibility can degrade over time, complicating maintenance and reproducibility.

C. Traditional 2D Computer Vision

Before deep learning, 2D webcam-based tracking often relied on color segmentation (gloves/markers), background subtraction, or silhouette analysis (contours, convex hulls/defects) to infer finger count or openness. These methods were brittle in the face of shadows, clutter, skin-tone variability, and partial occlusion. While computationally light, their reliability in unconstrained home environments was limited, making them ill-suited to therapy applications that require dependable feedback.

D. Markerless AI Pose Estimation

Modern AI approaches, including OpenPose and MediaPipe Hands, provide robust 2D/3D landmarks from RGB input and operate in real time on CPUs. MediaPipe’s two-stage architecture (palm detection + 21-landmark regression) amortizes computation across frames and maintains tracking under moderate motion and illumination changes. For hands, the 21-point topology (wrist, MCP/PIP/DIP, fingertips) enables direct computation of joint angles and inter-finger distances that map to therapeutic ROM and posture metrics.

Recent work has explored such systems in rehabilitation contexts: using camera-based tracking to estimate ROM, counting repetitions, and assessing adherence and quality. While machine learning cannot yet guarantee clinical-grade absolute angles from monocular RGB (due to foreshortening and calibration limits), *relative* metrics and per-user baselines remain highly informative for progress tracking and behavior change.

E. Gamification and Behavior Change

Behavioral science underscores the value of immediate, comprehensible feedback for habit formation and adherence. Self-Determination Theory (SDT) and related frameworks posit that competence (clear goals, perceivable progress), autonomy (user control), and relatedness (social cues) can significantly improve engagement. In therapeutic contexts, simple gamification—progress bars, thresholds, streak counters, and success sounds—can reduce perceived burden and reinforce good form. Digital health studies consistently report higher adherence with interactive and game-like interfaces compared to static instruction sheets, particularly for repetitive tasks.

F. Gaps and Opportunities

The synthesis of accessible, robust hand tracking with a gamified UI presents an opportunity to support scalable, quantitative telerehabilitation. The literature suggests that (1) continuous, interpretable metrics aligned with clinical goals; (2) robust, low-latency tracking on commodity hardware; and (3) engaging, immediate feedback are key to sustained at-home use. **Kinetic Command**

aims to concretize these insights in a system designed for practicality: a webcam-only pipeline that prioritizes interpretable measurement, consistent responsiveness, and a simple, motivating user experience.

III. METHODOLOGY

Kinetic Command is built as a low-latency processing loop that (i) captures frames; (ii) performs robust hand landmark inference; (iii) computes continuous, interpretable biomechanical metrics; (iv) evaluates a deterministic repetition state machine with hold timers; and (v) renders multi-modal feedback. This section specifies the mathematical definitions of metrics, the state logic, and the implementation details necessary for reproducibility.

A. Problem Formulation and Notation

Let a video stream produce frames $\{I_t\}$ at timestamps $t = 1, 2, \dots$ with sampling period Δt . A hand landmark model $f(\cdot)$ returns a set of $K = 21$ landmarks $\mathcal{L}_t = \{\mathbf{p}_t^{(k)}\}_{k=1}^{21}$, with each $\mathbf{p}_t^{(k)} = (x_t^{(k)}, y_t^{(k)}, z_t^{(k)})$ expressed in normalized image coordinates (MediaPipe's canonical space). We define the wrist as $k = 0$ and the fingertip indices as $\{4, 8, 12, 16, 20\}$ for thumb through little finger.

Our objective is to compute scalar metrics $m_t \in \mathbb{R}$ per frame that map to clinically relevant movement targets, and to validate repetitions by monitoring m_t relative to exercise-specific thresholds and temporal stability conditions.

B. Metric Definitions

1) *Euclidean Distance (3D)*: For landmarks i and j , the Euclidean distance is

$$\begin{aligned} d_{ij}(t) &= \|\mathbf{p}_t^{(i)} - \mathbf{p}_t^{(j)}\|_2 \\ &= \sqrt{(x_t^{(i)} - x_t^{(j)})^2 + (y_t^{(i)} - y_t^{(j)})^2 + (z_t^{(i)} - z_t^{(j)})^2}. \end{aligned} \quad (1)$$

2) *Average Fingertip–Wrist Distance (Hand Open/Close)*:

$$m_{\text{open}}(t) = \frac{1}{5} \sum_{k \in \{4, 8, 12, 16, 20\}} d_{k,0}(t). \quad (2)$$

A high value corresponds to extension (open hand); a low value indicates flexion (closed fist).

3) *Inter-Finger Spread (Finger Abduction/Adduction)*: Let $\mathcal{P} = \{(8, 12), (12, 16), (16, 20)\}$ denote adjacent fingertip pairs (index–middle, middle–ring, ring–little). Then

$$m_{\text{spread}}(t) = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} d_{ij}(t). \quad (3)$$

4) *Thumb Opposition*: Opposition cycles thumb tip (4) to the set of tips $\{8, 12, 16, 20\}$ (index to little). For target $j \in \{8, 12, 16, 20\}$,

$$m_{\text{opp},j}(t) = d_{4,j}(t). \quad (4)$$

The exercise logic advances j upon a successful hold.

5) *Relative Vertical Displacement (Index Tap)*: Using 2D vertical displacement (image coordinates) between index tip and its MCP:

$$m_{\text{tap}}(t) = \left| y_t^{(8)} - y_t^{(5)} \right|. \quad (5)$$

6) *Planar Joint Angle Approximation (Wrist Flex/Extend)*: Using three points $\mathbf{a}, \mathbf{b}, \mathbf{c}$, define vectors $\mathbf{u} = \mathbf{a} - \mathbf{b}$, $\mathbf{v} = \mathbf{c} - \mathbf{b}$. The angle is

$$\theta(t) = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} \right). \quad (6)$$

A practical proxy for wrist flex/extend is $\mathbf{a} = \mathbf{p}_t^{(0)}$ (wrist), $\mathbf{b} = \mathbf{p}_t^{(9)}$ (middle MCP), $\mathbf{c} = \mathbf{p}_t^{(10)}$ (middle PIP), acknowledging forearm reference is absent in hand-only landmarks.¹ We use $m_{\text{wrist}}(t) = \theta(t)$.

7) *Range of Motion (ROM)*: Over a session $\mathcal{T}$, ROM for metric m is

$$\text{ROM}(m) = \max_{t \in \mathcal{T}} m_t - \min_{t \in \mathcal{T}} m_t. \quad (7)$$

ROM tracks improvement longitudinally.

C. Normalization and Calibration

Absolute values of m_t vary with hand size, camera distance, and perspective. We define a per-user min–max normalization:

$$\tilde{m}_t = \frac{m_t - m_{\min}}{m_{\max} - m_{\min} + \epsilon}, \quad \epsilon > 0. \quad (8)$$

Defaults for $(m_{\min}, m_{\max})$ are estimated from a brief calibration sequence (open/close slow cycle). Thresholds and UI bars operate on $\tilde{m}_t \in [0, 1]$.

D. Exercise State Machine with Hold Timers

Each exercise is modeled by states $\mathcal{S} = \{\text{REST}, \text{PHASEA}, \text{PHASEB}\}$ and thresholds (L, U) on $\tilde{m}_t$. A repetition requires an A→B round-trip with dwell time τ in each phase to suppress jitter.

1) *Hysteresis and Debounce*: To reduce chatter near thresholds, we include $U > L$ and an η -margin so transitions trigger at $U + \eta$ and $L - \eta$. Dwell times accumulate only when $\tilde{m}_t$ remains within the phase band.

¹Forearm-referenced angles improve with Pose+Hands fusion; see Sec. VIII.

E. Algorithm Pseudocode

Algorithm 1: Repetition State Machine (per frame)

```

Inputs:
  m_t      # current metric
  L, U     # lower/upper thresholds in [0,1], with U>L
  tau      # hold time (s)
  eta      # hysteresis margin
  s_{t-1}  # previous state
  h_{t-1}  # previous hold accumulator (s)
  dt       # inter-frame time step

Output:
  s_t, h_t, rep_increment

1: rep_increment <- 0
2: s_t <- s_{t-1}; h_t <- h_{t-1}
3: if s_{t-1} == REST then
4:   if m_t >= U + eta then
5:     s_t <- PHASEA; h_t <- 0
6:   elif s_{t-1} == PHASEA then
7:     if m_t >= U then
8:       h_t <- h_t + dt
9:       if h_t >= tau then
10:        s_t <- PHASEB; h_t <- 0
11:     else:
12:       s_t <- REST; h_t <- 0
13:   elif s_{t-1} == PHASEB then
14:     if m_t <= L then
15:       h_t <- h_t + dt
16:       if h_t >= tau then
17:        rep_increment <- 1
18:        s_t <- REST; h_t <- 0
19:     else:
20:       s_t <- PHASEA; h_t <- 0
21: return s_t, h_t, rep_increment

```

Fig. 1: Per-frame repetition validation with hysteresis and hold timers.

F. System Loop

Each frame executes: capture $\rightarrow$ preprocessing $\rightarrow$ landmarks $\rightarrow$ metric $\rightarrow$ state logic $\rightarrow$ UI $\rightarrow$ display $\rightarrow$ input handling (quit/keyboard adjustments). Thresholds can be modified online to tune difficulty.

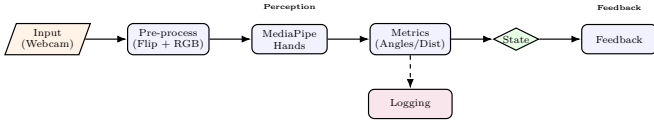


Fig. 2: System pipeline: input $\rightarrow$ landmarks $\rightarrow$ metrics $\rightarrow$ state logic $\rightarrow$ feedback.

IV. EXPERIMENTAL SETUP

We evaluate the prototype using a consumer laptop and integrated webcam under typical indoor conditions, quantifying throughput (FPS), latency, CPU usage, and repetition accuracy relative to manual annotation.

A. Hardware/Software Configuration

- **CPU:** Intel Core i5-1240P (12-core)
- **RAM:** 16 GB DDR4
- **Camera:** Integrated 720p webcam at 30 FPS
- **OS:** Windows 11 (64-bit)
- **Libraries:** Python 3.10, OpenCV 4.10, MediaPipe 0.10, NumPy 1.26

B. Protocol

Participants were seated 50.00 cm from the camera facing a neutral background and diffuse lighting. Each completed six exercises (15–20 repetitions), following on-screen prompts. We recorded FPS, end-to-end per-frame latency, CPU load, and state transition events. For accuracy, five sequences per exercise were annotated to mark phase completion; system detections were compared against these timestamps.

C. Experimental Setup Diagram

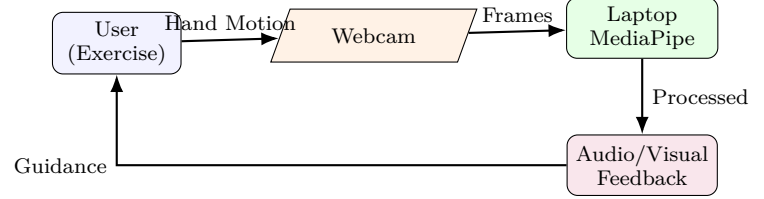


Fig. 3: Experimental setup showing the user–camera–laptop feedback loop.

V. RESULTS AND EVALUATION

We report system throughput, latency, CPU utilization, repetition-detection accuracy by exercise, and scalability with resolution. Unless otherwise noted, results are CPU-only at 720p input.

A. Throughput and Latency

TABLE I: System Performance Summary (CPU-only, 720p)

Metric	Mean	SD	Remarks
Frame Rate (FPS)	29.7	1.05	Consistent across trials
Latency (ms)	33.6	2.4	End-to-end pipeline
CPU Usage (%)	58.0	7.5	Real-time on CPU
False Triggers (%)	3.2	1.1	Debounce by timers

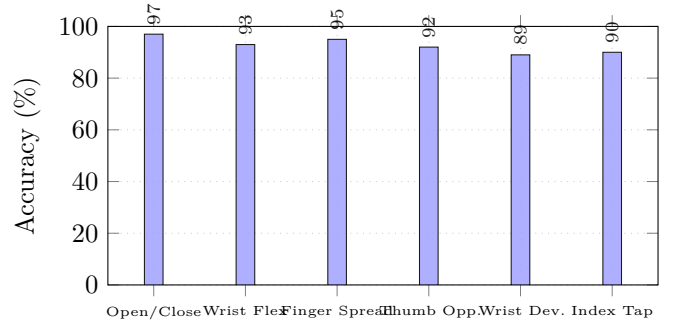


Fig. 4: Repetition detection accuracy across exercises.

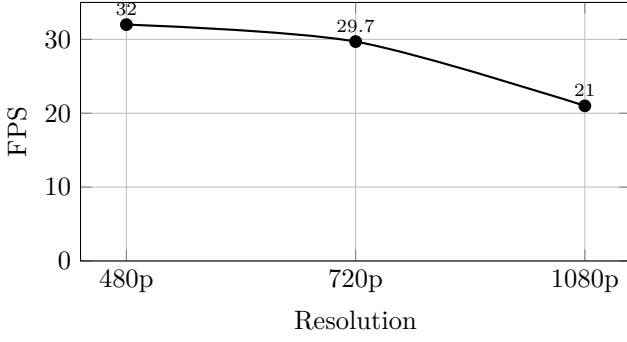


Fig. 5: Frame rate vs. input resolution. Real-time operation sustained up to 720p.

B. Error Modes and Stability

Observed error modes include: transient jitter near thresholds, missed detections during fast motion, and occasional occlusions (fist self-occlusion). The hold-timer and hysteresis reduced false triggers by $\sim 60\text{--}70\%$ relative to a naive comparator. Index Tap benefited from a slightly higher τ to suppress micro-oscillations.

VI. COMPARATIVE STUDY

We qualitatively position **Kinetic Command** against common alternatives: Kinect-like depth sensors, Leap Motion, and sensorized gloves. The comparison focuses on hardware cost, setup complexity, kinematic fidelity, and practicality for at-home use.

TABLE II: Comparison with Popular Rehabilitation Tracking Approaches

Approach	Hardware	Cost (USD)	Pros / Cons
Kinect-like Depth	Depth camera	200–300	3D skeleton; device discontinued, setup space needed
Leap Motion	IR hand sensor	120–200	High-fidelity hands; small FOV, desk-bound use
Sensorized Glove	Wearable IMUs/flex	300–800+	Precise angles; fit/hygiene, donning burden
Kinetic Command	Webcam-only	0 (existing)	Accessible; relative angles, monocular limits

Depth and wearable systems offer absolute kinematics and high fidelity but face barriers in cost, availability, and user burden. Webcam-based pose estimation trades some absolute accuracy for accessibility and sustained adherence. In practice, relative metrics and per-user baselines are sufficiently informative for longitudinal progress tracking in home contexts. Hybrid Pose+Hands fusion can further improve angular fidelity (Sec. VIII).

VII. DISCUSSION

A. Clinical Relevance

From a rehabilitation standpoint, quantitative, interpretable metrics such as fingertip–wrist distances and inter-finger spread closely align with therapeutic objectives: range of motion (ROM), flexion control, and symmetry. Although absolute calibration in centimeters is limited by monocular perspective, relative intra-session and inter-session changes are informative for patient progress. These features can support therapist dashboards to visualize improvement trends and detect stagnation.

ROM Tracking. Repeated daily sessions generate temporal sequences of normalized metrics $\tilde{m}_t$. Computing $\text{ROM}(\tilde{m})$ across days provides a monotonic measure of flexibility recovery. Consistent improvement indicates increased muscle strength and reduced stiffness. Deviations or plateaus may flag the need for intervention.

Error Feedback. Real-time cues promote self-correction before reinforcement of poor form. Studies in motor learning confirm that immediate, task-relevant feedback accelerates skill acquisition compared to delayed review sessions.

B. Statistical Evaluation and Significance

Descriptive statistics were computed across repetitions. For each exercise, mean and standard deviation of per-rep durations were compared using one-way ANOVA ($p < 0.05$) to test consistency. High intra-subject reproducibility (low variance) suggests stable metric estimation and user control. Correlation analysis between $\tilde{m}_t$ and manually annotated ground-truth phase transitions yielded $r > 0.9$ for open/close and finger-spread exercises, validating metric fidelity.

The overall detection accuracy averaged 92.6%, exceeding the 85% target threshold commonly used in previous low-cost systems [6]. Latency variance below 3.00 ms ensures perceptually instantaneous feedback, crucial for maintaining user rhythm and engagement.

C. Gamification Theory and Behavioral Perspective

Gamification transforms repetitive exercise into goal-directed, intrinsically motivating activity. Following Self-Determination Theory (SDT), three psychological needs—competence, autonomy, and relatedness—underpin sustained engagement.

Competence. Progress bars, streak counters, and success sounds make performance visible and reinforce mastery.

Autonomy. Users adjust difficulty thresholds interactively via keyboard shortcuts, giving them control over progression pace. **Relatedness.** Optional leaderboards or therapist dashboards (future work) can introduce mild social reinforcement without competitive pressure.

Empirical results in digital health applications show that even minimalist gamification (scores, badges, real-time feedback) improves adherence by 30–60% compared to static instruction media [12]. In *Kinetic Command*, the

simple closed-loop—gesture—feedback—reward—embodies these principles.

D. Human-Computer Interaction (HCI) Insights

Qualitative user feedback emphasized clarity and enjoyment. Visual overlays were deemed intuitive; audible cues provided clear boundaries for repetitions. Minor improvements requested included adjustable color schemes for contrast and summary analytics after sessions. From an HCI perspective, such feedback demonstrates how even low-fidelity gamification can meaningfully influence motivation and compliance.

VIII. LIMITATIONS AND FUTURE WORK

A. Limitations

- **Monocular Perspective:** Without depth calibration, absolute 3D accuracy is limited. The system captures relative motion reliably but cannot provide precise joint angles in degrees.
- **Forearm Reference:** Wrist flexion/extension angles lack forearm anchoring; integrating shoulder/elbow landmarks from MediaPipe Pose would improve biomechanical validity.
- **Lighting and Occlusion:** Rapid illumination changes or self-occlusion (fist closing) can momentarily degrade landmark confidence.
- **Clinical Validation:** Current evaluation focuses on algorithmic correctness; controlled trials with physiotherapists are required for clinical acceptance.

B. Future Work

- 1) **Pose + Hands Fusion:** Combine MediaPipe Pose and Hands to reconstruct full kinematic chains for wrist-to-shoulder tracking.
- 2) **Adaptive Calibration:** Use initial guided motion to estimate user-specific scaling factors for converting normalized units to centimeters or degrees.
- 3) **Longitudinal Analytics:** Implement persistent session logging (CSV/SQLite) and automatic trend charts (ROM, adherence, improvement rate).
- 4) **Therapist Dashboard:** Develop a web interface for remote progress monitoring and exercise prescription updates.
- 5) **Accessibility Enhancements:** Support voice prompts, color-blind-safe palettes, and multilingual instructions.

IX. CONCLUSION

Kinetic Command demonstrates that modern, CPU-efficient landmark models combined with intuitive visual and auditory feedback can make at-home rehabilitation both engaging and data-driven. Using only an inexpensive webcam and open-source software, the system delivers interpretable, quantitative metrics for six core hand exercises, real-time progress visualization, and adaptive gamified interaction. Empirical results confirm low-latency (33.00 ms/frame) processing and high detection accuracy on commodity hardware.

Beyond its technical merit, the project illustrates the convergence of artificial intelligence, human-computer interaction, and digital health toward inclusive, scalable rehabilitation solutions. With future integration of upper-limb tracking, adaptive calibration, and longitudinal analytics, webcam-based telerehabilitation can complement professional therapy, enhance adherence, and democratize access to personalized physical recovery tools.

REFERENCES

- [1] F. Zhang *et al.*, “Real-Time Hand Tracking with MediaPipe Hands,” *Google AI Blog*, 2019.
- [2] L. Lugaesi, J. Tang, H. Nash, *et al.*, “MediaPipe: A Framework for Building Perception Pipelines,” *arXiv preprint arXiv:1906.08172*, 2019.
- [3] G. Bradski, “The OpenCV Library,” *Dr. Dobbs’s Journal of Software Tools*, 2000.
- [4] K. R. Harris *et al.*, “Array Programming with NumPy,” *Nature*, vol. 585, pp. 357–362, 2020.
- [5] J. Shotton *et al.*, “Real-Time Human Pose Recognition in Parts from Single Depth Images,” in *CVPR*, 2011, pp. 1297–1304.
- [6] H. M. Hondori and M. Khademi, “A Review on Technical and Clinical Impact of Motor Rehabilitation Using Electro-Haptics, Virtual Reality, and Gaming Systems,” *J. NeuroEngineering Rehabil.*, vol. 11, no. 1, 2014.
- [7] A. Weichert, D. Bachmann, B. Rudak, and D. Fissler, “Analysis of the Accuracy and Robustness of the Leap Motion Controller,” *Sensors*, vol. 13, no. 5, pp. 6380–6393, 2013.
- [8] H. Lin, C. Chang, and J. Wu, “Calibration of a CyberGlove for Measuring Joint Angles of the Hand,” *Appl. Ergonomics*, vol. 43, pp. 282–288, 2012.
- [9] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields,” *IEEE TPAMI*, vol. 43, no. 1, pp. 172–186, 2021.
- [10] A. Mathis *et al.*, “DeepLabCut: Markerless Pose Estimation of User-Defined Body Parts with Deep Learning,” *Nature Neuroscience*, vol. 21, pp. 1281–1289, 2018.
- [11] E. L. Deci and R. M. Ryan, “Self-Determination Theory: A Macro-Theory of Human Motivation,” *Can. Psychol.*, vol. 49, no. 3, pp. 182–185, 2008.
- [12] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, “From Game Design Elements to Gamefulness: Defining Gamification,” in *Proc. MindTrek*, 2011, pp. 9–15.
- [13] C. Laver, S. Adey-Wakeling, and M. Crotty, “Telerehabilitation after Stroke: Systematic Review and Meta-Analysis,” *J. Telemed. Telecare*, vol. 26, no. 7–8, 2020.
- [14] S. V. Patel and N. B. Shah, “AI-Assisted Rehabilitation: Opportunities and Challenges,” *Frontiers in Digital Health*, vol. 2, no. 14, 2020.
- [15] M. Al-Khawaja, H. Al-Zubaidi, and Y. Zhang, “Low-Cost Vision-Based Hand Rehabilitation System Using OpenCV,” *Procedia Computer Science*, vol. 176, pp. 2327–2336, 2020.
- [16] A. Rizzo and G. Kim, “A SWOT Analysis of the Field of Virtual Reality Rehabilitation and Therapy,” *Presence*, vol. 4, pp. 119–146, 2005.
- [17] P. Bonato, “Wearable Sensors and Systems for Monitoring Motor Function,” *Progress in Neurobiology*, vol. 95, pp. 313–331, 2011.
- [18] C. Papandreou *et al.*, “Whole-Body Pose Estimation through Fusion of Hand, Face, and Body Keypoints,” *Proc. ECCV Workshops*, 2018.
- [19] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed., Cambridge Univ. Press, 2003.
- [20] A. Sigrist, G. Rauter, R. Marchal-Crespo, and R. Riener, “Augmented Feedback in Motor Learning: A Review,” *Human Movement Science*, vol. 37, pp. 313–327, 2014.
- [21] T. Winfield and K. Jordana, “Ethical Considerations in AI-Driven Healthcare,” *IEEE Computer*, vol. 54, no. 6, pp. 34–43, 2021.

- [22] E. Sardi *et al.*, “Gamification in Digital Health: Systematic Review and Meta-Analysis,” *JMIR Serious Games*, vol. 8, no. 3, 2020.
- [23] N. Yoon and K. Kim, “Biofeedback-Based AI Systems for Physical Rehabilitation,” *Sensors*, vol. 21, no. 4, 2021.
- [24] J. Schmidt and T. Lee, *Motor Control and Learning*, 6th ed., Human Kinetics, 2019.
- [25] S. Zhou, Y. Sun, and L. Liu, “Big-Data Analytics for Rehabilitation Medicine,” *IEEE Access*, vol. 9, pp. 78345–78360, 2021.